🔐 Semptify Server Setup

Credentials & External Services Configuration Workbook

Version 5.0.0 | Generated for New Server Deployment

📋 How to Use This Workbook

This document guides you through obtaining and configuring ALL external services needed for a fully functional Semptify server. Fill in each section as you obtain credentials from external providers.

Print this document or use digital PDF tools with form fields to complete.

📝 Setup Checklist

- [ ] **Cloud Storage (OAuth)** - Google Drive, Dropbox, OneDrive
- [ ] **System Storage (R2)** - Cloudflare R2 for production file storage
- [ ] **AI APIs** - OpenAI, Anthropic, Groq, Azure OpenAI, Google Gemini
- [ ] **Database** - PostgreSQL connection string (production)
- [ ] **Security Keys** - Secret key generation and admin PIN
- [ ] **Optional APIs** - GitHub, MN County APIs, News APIs

☁️ Google Drive OAuth Setup

Step-by-Step Guide:

1. Go to **https://console.cloud.google.com/**
2. Click "Select a Project" → "New Project"
3. Name: `Semptify` → Create
4. Search for "Google Drive API" → Click → Enable
5. Go to "Credentials" (left sidebar)
6. Click "Create Credentials" → "OAuth client ID"
7. Choose "Web application"
8. Add Authorized JavaScript origins: `http://localhost:8000`
9. Add Authorized redirect URIs (see below)
10. Click "Create" → Copy Client ID and Client Secret

Redirect URIs:

```
http://localhost:8000/storage/oauth/google/callback

http://yourdomain.com/storage/oauth/google/callback
```

Google Drive Client ID *

144818423056-tu9evppusv3r3933amc3ant8hu47uouh.apps.googleusercontent.co

Google Drive Client Secret *

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

☑ Credentials obtained

📦 Dropbox OAuth Setup

Step-by-Step Guide:

1. Go to **https://www.dropbox.com/developers/apps**
2. Click "Create app"
3. Choose:
   - API: Scoped access
   - Type: Full Dropbox
4. App name: `Semptify`
5. Go to "Settings" tab → OAuth 2
6. Add Redirect URIs (see below)
7. Go to "Permissions" → Enable `files.content.read`, `files.content.write`, `files.metadata.read`
8. Copy App Key and App Secret from "Settings"

Redirect URIs:

```
http://localhost:8000/storage/oauth/dropbox/callback

http://yourdomain.com/storage/oauth/dropbox/callback
```

Dropbox App Key *

`0d9rqwu5cwkgssu`

Dropbox App Secret *

`vwmui47qhrijkdh`

☑ Credentials obtained

☁️ OneDrive / Microsoft OAuth Setup

Step-by-Step Guide:

1. Go to **https://portal.azure.com/**
2. Search for "App registrations" → Click
3. Click "New registration"
4. Name: `Semptify`
5. Supported account types: Accounts in any organizational directory
6. Redirect URI: Select "Web" and add URI (see below)
7. Register → Copy "Application (client) ID"
8. Go to "Certificates & secrets" → New client secret
9. Copy the secret value (not ID)
10. Go to "API permissions" → Add "Microsoft Graph" → "Files.ReadWrite.All"

Redirect URIs:

```
http://localhost:8000/storage/oauth/onedrive/callback

http://yourdomain.com/storage/oauth/onedrive/callback
```

OneDrive / Microsoft Client ID *

a3a1b348-0756-443e-93bb-df2bf9d96a94

OneDrive / Microsoft Client Secret *

a8c2909d-62ae-4b1f-a0a8-2c4a69f5be10

☑ Credentials obtained

🔒 Cloudflare R2 Setup (Production System Storage)

⚠️ **Important:** R2 is required for production deployments and stateless server instances. Local development can use SQLite with in-memory storage.

Step-by-Step Guide:

1. Go to **https://dash.cloudflare.com/**
2. Log in → Select account
3. Left sidebar → "R2" → "Create bucket"
4. Bucket name: `semptify-server-prod`
5. Leave location as "Default"
6. Create → Save bucket name and region
7. Go to "Settings" → Copy endpoint URL
8. Left sidebar → "API Tokens" → Create API token
9. Select "R2" scope → "All buckets" permissions → Generate
10. Copy Account ID, Access Key ID, Secret Access Key

R2 Configuration:

R2 Account ID *

```
be2a39cd3624261169fa8e800d75923f
```

R2 Access Key ID *

```
11587fec6331ea889c3e02a819c784bf
```

R2 Secret Access Key *

```
57a6a42b14139ce9cfc915a13c6659e14e94c63851519f6739cd028957f6e20f
```

R2 Bucket Name *

```
semptify/
```

R2 Endpoint *

```
https://be2a39cd3624261169fa8e800d75923f.r2.cloudflarestorage.com
```

R2 API Token (for programmatic access) Optional

```
Long-lived token if needed for automation
```

- [x] R2 bucket created
- [x] API credentials obtained

🤖 AI Providers Configuration

Primary AI Provider (Choose ONE):

Recommended: Anthropic Claude (most reliable for legal analysis)

Anthropic Claude * RECOMMENDED

1. Go to **https://console.anthropic.com/**
2. Sign up or log in
3. Go to "Account" → "API keys"
4. Click "Create key"
5. Copy the full API key

Anthropic API Key *

```
sk-ant-...
```

☐ Anthropic configured

OpenAI (ChatGPT, GPT-4) Optional

1. Go to **https://platform.openai.com/**
2. Log in → Account → API keys
3. Create new secret key
4. Copy immediately (can only be viewed once)

OpenAI API Key Optional

```
sk-proj-...
```

- [ ] OpenAI configured

Groq (Fast inference) Optional

1. Go to **https://console.groq.com/**
2. Create account → API keys
3. Create key and copy

Groq API Key Optional

```
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
```

- [x] Groq configured

Google Gemini Optional

1. Go to **https://aistudio.google.com/app/apikey**
2. Click "Create API key"
3. Copy the key

Google Gemini API Key Optional

```
AQ.Ab8RN6Jgi769gap7gB2XtllAPNGaJAvHJJzIe8ZYyMe2GLcESA
```

- [x] Google Gemini configured

Azure OpenAI Optional - Enterprise

Azure OpenAI API Key Optional

```
Azure API key
```

Azure OpenAI Endpoint Optional

```
https://*.openai.azure.com/
```

- [ ] Azure OpenAI configured

🗄️ Database Configuration

Development: SQLite (local, automatic)
Production: PostgreSQL (recommended)

For Development (SQLite) - Automatic

SQLite database automatically created at: `./semptify.db`
No configuration needed for local development.

For Production (PostgreSQL)

Option A: Managed Service (Recommended)

- Use **AWS RDS**, **Azure Database**, or **Heroku Postgres**
- Create database named `semptify`
- Create user and set password
- Note the connection details

Option B: Self-Hosted

- Install PostgreSQL 13+
- Create database: `createdb semptify`
- Create user and password

Database Type *

```
Choose: sqlite (dev) or postgresql (prod)
```

DATABASE_URL *

```
postgresql+asyncpg://user:password@host:5432/semptify
```

```
# SQLite Example (Development)
DATABASE_URL=sqlite+aiosqlite:///./semptify.db # PostgreSQL Example
```

```
(Production)
DATABASE_URL=postgresql+asyncpg://semptify_user:secure_password@db.exa
```

- [ ] Database created
- [ ] Connection string tested

🔐 Security Configuration

Generate Secret Key

Run in terminal:

```
python -c "import secrets; print(secrets.token_urlsafe(64))"
```

Copy the output and paste below

SECRET_KEY *

```
Output from secrets.token_urlsafe(64)
```

Admin PIN (for emergency access)

Create a strong numeric PIN for administrative overrides

ADMIN_PIN *

```
041495
```

Security Mode

SECURITY_MODE=open → No OAuth required (development)
SECURITY_MODE=enforced → OAuth required (production)

SECURITY_MODE *

```
open (dev) or enforced (production)
```

- [ ] Secret key generated

- ☑ Admin PIN set

🔗 Optional External APIs

GitHub Token (for commit history integration) Optional

1. Go to GitHub Settings → Developer settings → Personal access tokens
2. Generate new token with `repo` scope

GITHUB_TOKEN Optional

```
ghp_...
```

Minnesota County APIs (for property research) Optional

County Assessor Base URL Optional

```
https://gis.hennepin.us/property/api/v1
```

County Assessor API Key Optional

County Recorder Base URL Optional

```
https://www.hennepin.us/recorder/api/v2
```

NewsAPI (for tenant issue tracking) Optional

1. Go to **https://newsapi.org/**
2. Sign up → Get API key

NEWS_API_KEY Optional

```
8bff1462c8a24e9ab946c7934449cba0
```

.env File Template

Copy and fill in this template with your credentials:

```
# SECURITY SECRET_KEY=_____ ADMIN_PIN=_____
SECURITY_MODE=enforced # DATABASE
DATABASE_URL=postgresql+asyncpg://user:pass@host:5432/semptify #
CLOUD STORAGE (OAUTH) GOOGLE_DRIVE_CLIENT_ID=_____
GOOGLE_DRIVE_CLIENT_SECRET=_____
DROPBOX_APP_KEY=_____
DROPBOX_APP_SECRET=_____
ONEDRIVE_CLIENT_ID=_____
ONEDRIVE_CLIENT_SECRET=_____ # CLOUDFLARE R2
(PRODUCTION) R2_ACCOUNT_ID=_____
R2_ACCESS_KEY_ID=_____
R2_SECRET_ACCESS_KEY=_____ R2_BUCKET_NAME=semptify-
server-prod R2_ENDPOINT=_____ # AI PROVIDERS
ANTHROPIC_API_KEY=_____ OPENAI_API_KEY=_____
GROQ_API_KEY=_____ GEMINI_API_KEY=_____
AZURE_OPENAI_API_KEY=_____
AZURE_OPENAI_ENDPOINT=_____ # OPTIONAL
GITHUB_TOKEN=_____
```

✅ Final Deployment Checklist

Before Going Live:

- [ ] All OAuth credentials configured (Google, Dropbox, OneDrive)
- [ ] R2 bucket created and credentials tested
- [ ] At least one AI provider configured (Anthropic recommended)
- [ ] PostgreSQL database connection tested
- [ ] Secret key generated and stored securely
- [ ] Admin PIN set (change from default)
- [ ] SECURITY_MODE set to "enforced"
- [ ] .env file created with all credentials
- [ ] .env file added to .gitignore (DO NOT COMMIT)
- [ ] Server started and tested: http://localhost:8000/onboarding
- [ ] SSL/TLS certificate configured (production only)
- [ ] Firewall rules configured
- [ ] Monitoring/logging set up
- [ ] Backups scheduled

⚠️ Security Reminders

- Never commit .env file to version control
- Use strong, unique passwords for all services
- Rotate API keys regularly
- Use environment variables in production, not hardcoded values
- Enable 2FA on all external service accounts
- Store credentials in a secure vault (1Password, LastPass, etc.)
- Back up SECRET_KEY and ADMIN_PIN in a secure location
- Monitor API usage and set billing alerts

📞 Support & Resources

Documentation

- **Semptify GitHub:** https://github.com/Semptify/Semptify-FastAPI
- **PRODUCTION.md:** Detailed deployment guide
- **API Docs:** http://your-domain:8000/docs

Helpful Links

Service	Setup URL
Google Cloud Console	https://console.cloud.google.com/
Dropbox Developers	https://www.dropbox.com/developers/apps
Azure Portal	https://portal.azure.com/
Cloudflare R2	https://dash.cloudflare.com/
Anthropic Console	https://console.anthropic.com/
OpenAI Platform	https://platform.openai.com/

> ✅ **Congratulations!** Once you've filled in all sections and deployed, your Semptify server is ready to protect tenant rights. Thank you for deploying this mission-critical tool.